

VoltEdge VE-9

Synchronous Buck Regulator - Datasheet Rev A

The VoltEdge VE-9 is a synchronous buck converter targeting the same 3.3 V logic rails as traditional linear parts, but at substantially higher efficiency. It accepts 4.5 V to 36 V input and delivers up to 3 A continuous at 94 percent peak efficiency, switching at 500 kHz.

Unlike a linear regulator, the VE-9 injects switching ripple onto the output rail: 35 mV peak-to-peak typical at full load. Noise-sensitive analog sections may require post-regulation or additional filtering.

Table 1: Electrical Characteristics

Parameter	Value
Input voltage range	4.5 V to 36 V
Output voltage	3.3 V adjustable
Continuous output current	3 A
Peak efficiency	94 percent
Switching frequency	500 kHz
Output ripple at full load	35 mV pp
Thermal shutdown threshold	165 C
Quiescent current	12 uA

Reliability and Supply

The VE-9 is qualified to AEC-Q100 Grade 2 only, and is not rated for the -40 C to +125 C ambient range required by Grade 1 designs. Sustained operation above +105 C ambient is outside the datasheet envelope.

The VE-9 entered volume production in Q1 and carries a 5-year supply commitment. Lead time is currently 22 weeks.